Molecular Properties through Coupled Cluster Theory

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Molecular Properties

What are they?

$$\hat{H}_0 \tag{1}$$

Unperturbed Hamiltonian

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$$\hat{H}_0 + V \tag{1}$$

Perturbation

Unperturbed Hamiltonian

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Perturbation

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$$\hat{H}(\varepsilon) = \hat{H}_0 + V(\varepsilon) \tag{1}$$

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- Perturbed Schrödinger equation:

$$\hat{H}(\varepsilon)\,|\Psi\rangle = E(\varepsilon)\,|\Psi\rangle \tag{2}$$

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- Perturbed Schrödinger equation:

$$\begin{aligned} \hat{H}(\varepsilon)\,|\Psi\rangle &= E(\varepsilon)\,|\Psi\rangle \\ E(\varepsilon) &= E^{[0]} + \varepsilon^T E^{[1]} + \frac{1}{2}\varepsilon^T E^{[2]}\varepsilon + \ldots \end{aligned} \tag{2}$$

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Time Independent Molecular Properties

$$\underbrace{E^{[1]} = \left.\frac{dE(\varepsilon)}{d\varepsilon}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}}}_{\text{First order}}, \quad \underbrace{E^{[2]} = \left.\frac{d^2E(\varepsilon)}{d\varepsilon^2}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}}}_{\text{Second order}}, \quad \cdots \quad , \underbrace{E^{[n]} = \left.\frac{d^nE(\varepsilon)}{d\varepsilon^n}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}}}_{\text{n-th order}}$$

Molecular Properties

Approximate wave functions

- Approximate wave functions: $|\Phi(\boldsymbol{\lambda})\rangle \Rightarrow E \equiv E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})$.

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- **Variational Perturbation Theory**: the energy is stationary for any perturbation strength.

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- The first order properties:

Response term

$$\left.\frac{dE(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left.\frac{\partial E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} + \left.\frac{\partial E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu} \frac{d\lambda_\mu}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{4}$$

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- The second order properties:

$$\left.\frac{d^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{d\varepsilon_A d\varepsilon_B}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left[\frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \varepsilon_A \partial \varepsilon_B} + \frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu \partial \varepsilon_A} \frac{d\lambda_\mu}{d\varepsilon_B}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{5}$$

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  - We need the first order of the **response of the wave function**.

$$\left.\frac{d}{d\varepsilon_B}\left(\frac{\partial E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu}\right)\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = 0 \Rightarrow \frac{d\lambda_\mu}{d\varepsilon_B} = \left[-\frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \varepsilon_B \partial \lambda_\mu}\left(\frac{\partial^2 E(\boldsymbol{\lambda}, \boldsymbol{\varepsilon})}{\partial \lambda_\mu^2}\right)^{-1}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{6}$$

Coupled Cluster Theory

A non complete introduction

- In coupled-cluster theory we use a non-unitary exponential parametrization:

Exponential Cluster operator

$$|\Psi_{\mathrm{CC}}\rangle = e^{\hat{T}} |\Phi_0\rangle \tag{7}$$

CC wave function

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CC wave function — Reference — Amplitudes — Excitation operators

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- By inserting the exponential ansatz into the Schrödinger equation, we obtain the energy:

$$\hat{H} |\Psi_{\mathrm{CC}}\rangle = E_{\mathrm{CC}} |\Psi_{\mathrm{CC}}\rangle \Rightarrow E_{\mathrm{CC}} = \frac{\langle \Psi_{\mathrm{CC}} | \hat{H} | \Psi_{\mathrm{CC}} \rangle}{\langle \Psi_{\mathrm{CC}} | \Psi_{\mathrm{CC}} \rangle} \tag{8}$$

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CC wave function — Reference — Amplitudes — Excitation operators

- In CC community we use the **similarity-transformed Hamiltonian**:

$$\hat{\overline{H}} \, |\Phi_0\rangle = E_{\mathrm{CC}} \, |\Phi_0\rangle \, , \quad \hat{\overline{H}} = e^{-\hat{T}} \hat{H} e^{\hat{T}} \tag{8}$$

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$$\hat{\overline{H}} = e^{-\hat{T}} \hat{H} e^{\hat{T}} = \hat{H} + [\hat{H}, \hat{T}] + \frac{1}{2!}[[\hat{H}, \hat{T}], \hat{T}] + \ldots \Rightarrow \text{Termination at the fourth term}$$

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$$= (\hat{H} e^{\hat{T}})_{\mathcal{C}} = \hat{H} + (\hat{H}\hat{T})_{\mathcal{C}} + \frac{1}{2!}((\hat{H}\hat{T})_{\mathcal{C}}\hat{T})_{\mathcal{C}} + \ldots \Rightarrow \text{The connected cluster theorem}$$

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- Projecting

$$\langle \Phi_0 | \quad : \quad \langle \Phi_0 | \hat{\overline{H}} | \Phi_0 \rangle = E_{\rm CC} \leftarrow \text{Energy (not an upper bound)} \tag{9}$$

$$\langle \Phi_\mu | \quad : \quad \langle \Phi_\mu | \hat{\overline{H}} | \Phi_0 \rangle = 0 \leftarrow \text{Amplitudes} \tag{10}$$

- We have

$$\left.\frac{dE_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=0} = \left[\frac{\partial E_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{\partial \varepsilon_A} + \frac{\partial E_{\mathrm{CC}}(\boldsymbol{t},\boldsymbol{\varepsilon})}{\partial t_\nu}\frac{dt_\nu}{d\varepsilon_A}\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{11}$$

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  - So much effort to just have the gradient... Can we do better?

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  - **Yes!**

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T.Helgaker and P.Jørgensen. *Theor. Chim. Acta*, 75:111, 1989

- We introduce the CC Lagrangian:

$$L_{\mathrm{CC}}(\boldsymbol{t}, \bar{\boldsymbol{t}}, \boldsymbol{\varepsilon}) = E_{\mathrm{CC}}(\boldsymbol{t}, \boldsymbol{\varepsilon}) + \sum_{\mu} \bar{t}_{\mu} \langle \Phi_{\mu} | \hat{\bar{H}} | \Phi_0 \rangle \tag{12}$$

Multipliers

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$$L_{\mathrm{CC}}(\boldsymbol{t}, \bar{\boldsymbol{t}}, \boldsymbol{\varepsilon}) = \langle \Phi_0 | (1 + \hat{\Lambda}) \hat{\overline{H}} | \Phi_0 \rangle \,, \quad \hat{\Lambda} = \sum_{\mu} \bar{t}_{\mu} \, \hat{\tau}_{\mu}^{\dagger} \tag{12}$$

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- Differentiating it

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  - We impose the stationarity of the Lagrangian with respect to t_ν and $\bar{t}_\nu$:

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  - We conclude that:

$$\left.\frac{dL_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left.\frac{\partial L_{\mathrm{CC}}(\boldsymbol{t},\bar{\boldsymbol{t}},\boldsymbol{\varepsilon})}{\partial\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \left[\langle\Phi_0|(1+\hat{\Lambda})\frac{\partial\hat{\overline{H}}}{\partial\varepsilon_A}|\Phi_0\rangle\right]_{\boldsymbol{\varepsilon}=\mathbf{0}} \tag{16}$$

Molecular Properties through Coupled Cluster Theory

Lambda equations

- The equations for the multipliers

$$\langle \Phi_0 | (1 + \hat{\Lambda}) [\overline{\hat{H}}, \hat{\tau}_\nu] | \Phi_0 \rangle = 0 \tag{17}$$

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- After some manipulations we get

$$\underbrace{\langle \Phi_0 | \overline{\hat{H}} | \Phi_\nu \rangle + \langle \Phi_0 | [\hat{\Lambda}, \overline{\hat{H}}] | \Phi_\nu \rangle}_{\text{Connected part}} + \sum_{k \neq 0} \underbrace{\langle \Phi_0 | \overline{\hat{H}} | \Phi_k \rangle \langle \Phi_k | \hat{\Lambda} | \Phi_\nu \rangle}_{\text{Disconnected part}} = 0 \tag{18}$$



(1) (2)

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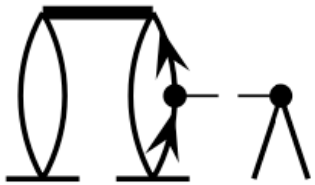
- As an example, for Λ-CCSD we have

$$\begin{aligned} &\langle \Phi_0 | \overline{\hat{H}} | \Phi_i^a \rangle_{\mathcal{C}} + \langle \Phi_0 | (\hat{\Lambda} \overline{\hat{H}})_{\mathcal{C}} | \Phi_i^a \rangle_{\mathcal{C}} = 0 \\ &\langle \Phi_0 | \overline{\hat{H}} | \Phi_{ij}^{ab} \rangle_{\mathcal{C}} + \langle \Phi_0 | (\hat{\Lambda} \overline{\hat{H}})_{\mathcal{C}} | \Phi_{ij}^{ab} \rangle_{\mathcal{C}} + \sum_{k,c} \langle \Phi_0 | \overline{\hat{H}} | \Phi_k^c \rangle \langle \Phi_k^c | \hat{\Lambda} | \Phi_{ij}^{ab} \rangle = 0 \end{aligned} \tag{19}$$

Molecular Properties through Coupled Cluster Theory

Some precautions

1. Each $\hat{T}$ must have a direct connection (contraction) to the Hamiltonian.

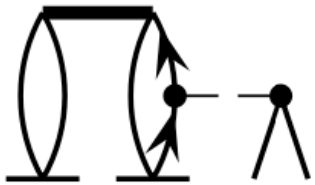


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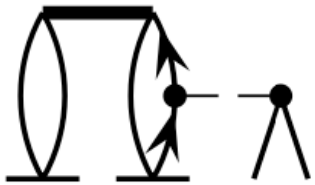
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- Solution: remove diagrams associated with terms that have t amplitudes not contracted with the Hamiltonian.

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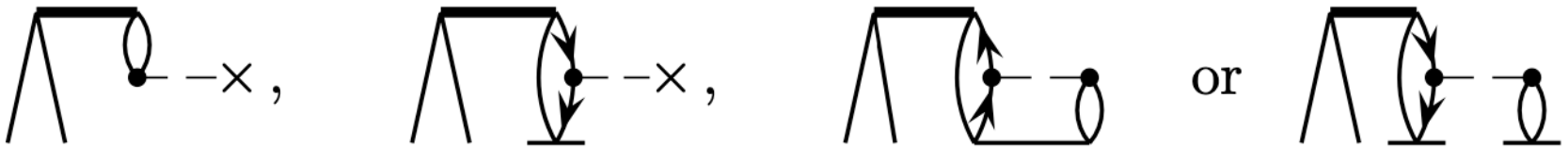
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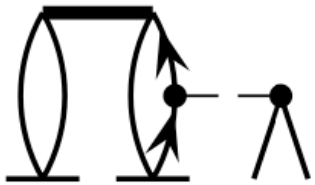


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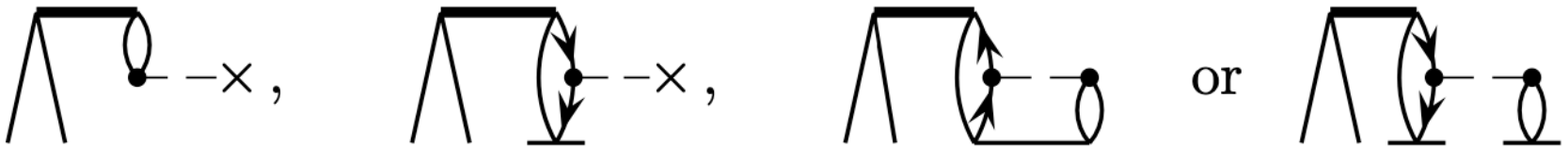
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- Solution: remove the diagrams associated with terms that have all indices of the part of the Hamiltonian contracted while there is a pair of lambda indices not contracted.

Molecular Properties through Coupled Cluster Theory

Expectation values

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- The first order CC molecular property will be

$$\left.\frac{dL_{\mathrm{CC}}}{d\varepsilon_A}\right|_{\boldsymbol{\varepsilon}=\mathbf{0}} = \sum_{pq} h_{pq;A} \gamma_{pq}^{\mathrm{CC}} \tag{24}$$

Conclusions and Future perspectives

1. CC development
   - Based on the **tenpi** toolchain. We are still working on:
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Conclusions and Future perspectives

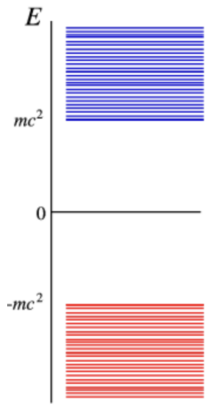
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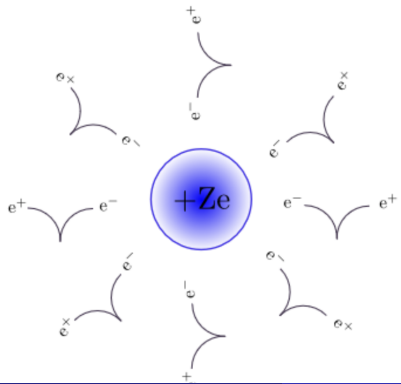
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   - Core properties: where **QED-effects** may play an important role.





HAMP-vQED

Highly Accurate Molecular Properties using variational Quantum Electrodynamics

HAMP-vQED team



Figure 1: Jan Brandejs, Gabriele Fabbro, Trond Saue, Ryan Benazzouk and Dávid Ferenc.